



Figure 1: A very simple IVR

and provide the information in a recorded voice or a text-to-speech (TTS) engine.

A typical IVR, as in a company's welcome script, is static—including the voice recordings. Dynamicity can be brought in with the integration of the following components.

- **Database:** The IVR script can query a database as well as store information into a database. Based on the data stored, the flow can be controlled or different information can be presented. The account status of a particular customer can be presented, after querying the database.
- **TTS (text-to-speech):** TTS engines help to read out the texts or numbers obtained from the database.
- **Voice recognition:** It is cumbersome for customers to type in all the numbers requested by the IVR, especially in the era of mobile phones. In the case of desktop phones, the process is manageable, as the number pad is in front of the user on the desk and only the headset is near the ear. With voice recognition, users can voice their choices, and the voice recognition engine will translate the voice to commands or information and control the flow.

As the world moves towards increased automation, the trend is to deploy more and more IVRs for all types of applications.

Call centre diallers

A dialler is another strong application based on Asterisk. To understand how it works, let's imagine a call centre in which multiple agents are seated, dialling thousands of customers. Typically, the team lead will provide them with a list of numbers and the agents will try to call each one of them. Most of the numbers will be non-reachable, busy or may not be picked up. Normally, a call centre agent needs to attempt at least three numbers to make one successful call. Assume that each attempt takes 30 seconds, and the successful call needs 60 seconds of talk time.

Total time for one call	=	(3×30) + 60 = 150 seconds
Total talk time	=	60 seconds
Efficiency	=	60/150 = 40 per cent

It's clear that 60 per cent of the agent's time is being wasted in non-productive work. For a call centre with 100 agents, 480 man hours are wasted every day, assuming an eight-hour day. With the average salary of an agent being Rs 10,000 per month, this amounts to a productivity loss of Rs 600,000 every month.

The dialler helps you save this huge wastage. How? The numbers can be uploaded into the dialler database and the dialler software will make the calls. The system automatically detects conditions like 'not reachable', 'busy', 'network congestion', 'not answering', etc. In case the call is established, the line is connected to a live agent.

Automatic answering machine detection is also included to avoid agents being connected to answering machines. If there are five agents, the system will send out 20 calls, so that the idle time for agents is reduced. The key is to find the optimum ratio, so that less manpower is wasted waiting for connected calls. The optimum ratio depends on multiple factors like lead quality, the day of the week, the time of the day, etc. Dialler software also enables redialling the unsuccessful calls, based on the reason the call failed on the first attempt.

Dialler software provide a lot of reports, which help the management to efficiently run the call centre. Agent-wise reports give an overview of the time utilised by each agent, including talk time, wait time and pauses. Similarly, we can also get data like the success rate with each list of numbers, average call duration and disposition status. Vicidial is the most popular open source dialler, which can be downloaded from vicidial.org.

ACD (automatic call distribution)

While diallers are used in outbound call centres, inbound call centres use ACD systems. The agents are connected to the ACD and segregated into different groups. The incoming call can be distributed among these groups of agents in multiple ways:

- **Language:** A simple IVR may help the customer select the preferred language and based on the selection, the call is forwarded to the respective language group.
- **Skills:** The agents are divided according to their skillsets, and the calls routed based on the skillset requested.
- **SLA:** The grouping of agents will be based on their experience levels and the routing can happen based on the SLAs (service level agreement), like platinum, gold or silver.
- **Region:** The caller's region can be ascertained based on the caller ID and routed accordingly.
- **Demography:** Callers may identify themselves and the routing can happen depending on their demographic attributes.

Similar to outbound calling, in inbound calling also, multiple reports are generated like agents' occupancy, call waiting times, and so on.